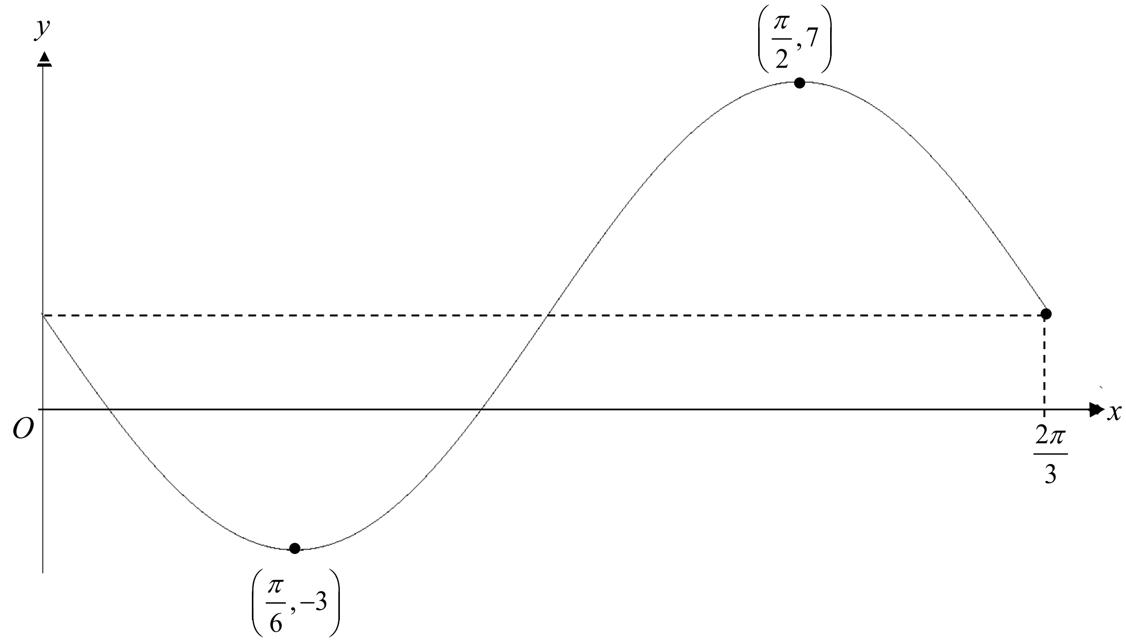
8.



The diagram shows part of the graph of $y = a\sin bx + c$ for $0 \le x \le \frac{2\pi}{3}$ radians.

The curve has a minimum point at $\left(\frac{\pi}{6}, -3\right)$ and a maximum point at $\left(\frac{\pi}{2}, 7\right)$.

(a) Explain why $c = 2$. [2]

(b) Explain why $b = 3$. [2]

(c) Hence find the equation of the curve. [2]

9. **(a)** Using $\sin 3x = \sin(2x + x)$, prove the identity,

$$\sin 3x + \sin x = 4\sin x\cos^2 x\,.$$ [3]

(b) Find the values of x for which $\sin 3x - 2\sin x = 0$, where $-\pi \leq x \leq \dfrac{\pi}{2}$. [5]

11. **(a)** Given $y = \dfrac{2x-3}{x^2+1}$, show that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{m(1+3x-x^2)}{(x^2+1)^2}$, where m is a constant.

State the value of m. [4]